radiation poisoning after mishandling bare plutonium cores in 1945 and 1946.[4] However, while American and foreign visitors had access to North Korea's Yongbyon nuclear reactor, plutonium reprocessing plant, and nuclear centrifuges for most of the period from 1995 through 2009, none have seen the actual factories where the North's bombs are ostensibly made, and thus few informed guesses can be made about North Korea's safety protocols in its nuclear production process. Any accident inside these confines would be a worrisome signal that Pyongyang was sacrificing safety for results.[5]

While we may never know about any North Korean accidents in laboratories or assembly plants, an accident like the September 1957 explosion at a nuclear waste processing and storage plant in Chelyabinsk, Russia, potentially could pose the first type of crisis the world may face with Pyongyang's nuclear program. Depending on its size, location, and prevailing wind patterns, a major release of nuclear material could wind up spreading radioactivity to South Korea, China, or Japan. (Figure 1 shows the locations of North Korea's nuclear facilities.) While the regional response would be focused on preventing civilian contamination, it would cause a diplomatic crisis. In response, Pyongyang might well try to deflect blame, claiming that sabotage was the cause, and even using that manufactured excuse to "retaliate" in turn.

The danger of potentially lax standards is compounded by normal wear and tear on nuclear systems. It's not surprising that the more nuclear systems are used, such as flying bombers, sailing submarines, or moving around missiles, the more likely accidents are to occur. During the Cold War, the relentless pace of constant American nuclear alerts led to numerous mishaps. In Operation Chrome Dome, for example, US Air Force B-52s carrying thermonuclear bombs were kept constantly in the air, flying to predetermined points around the Soviet Union, for eight full years, from 1960 to 1968. During that period, five major accidents occurred, ultimately leading to the cancellation of the program.

Nuclear weapons accidents such as these, called "broken arrows," nearly turned into catastrophe more than once. In late January 1961, a B-52 participating in Operation Chrome Dome flying out of Goldsboro,